

CCC_Quantum_Simulation:

Rigorous Quantum Algorithms for Conformal Cyclic Cosmology Subsystems

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Abstract

We present a comprehensive blueprint for quantum simulation of Conformal Cyclic Cosmology (CCC) subsystems. Our framework maps CCC's conformal geometry and massless field propagation to quantum-simulable operators, implements discretized Hamiltonians on quantum hardware, and validates transport of Hawking-point-like pulses across aeon boundaries. We develop three complementary Hamiltonian simulation approaches—Trotter–Suzuki decomposition, Linear Combination of Unitaries (LCU) with Quantum Signal Processing (QSP), and spectral methods via Quantum Fourier Transform—with explicit error bounds and gate counts. Detailed resource estimates demonstrate feasibility on near-term and fault-tolerant quantum devices. We provide fully worked examples in 1+1D and 2+1D, including explicit circuit constructions, angular radius calculations, and validation against exact solutions. This document serves as a rigorous, self-contained reference for researchers and engineers implementing quantum simulations of conformally invariant cosmological dynamics.

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1 Introduction

Conformal Cyclic Cosmology (CCC), proposed by Penrose [1], posits that the universe undergoes an infinite sequence of *aeons*, each beginning with a Big Bang and ending in a conformally smooth future infinity. The key physical insight is that at late times, when the universe becomes dominated by massless particles (after all massive particles decay), conformal invariance emerges, allowing a smooth transition to the next aeon’s Big Bang hypersurface.

1.1 CCC Assumptions and Physical Content

The fundamental assumptions of CCC are:

1. **Conformal continuation:** The future conformal boundary $\mathcal{I}^+$ of one aeon smoothly matches the initial Big Bang hypersurface $\mathcal{B}$ of the next aeon through a conformal rescaling.
2. **Mass extinction:** All massive particles eventually decay, leaving only conformally invariant massless fields.
3. **Information transfer:** Physical information, particularly from final-stage black hole evaporation (Hawking radiation [2]), propagates across the crossover surface $\Sigma = \mathcal{I}^+ \equiv \mathcal{B}$.

1.2 Simulation Scope

This document focuses on quantum simulation of the following CCC-relevant subsystems:

- **Conformally invariant massless field dynamics:** Scalar and electromagnetic fields propagating in conformally flat backgrounds.
- **Null congruence transport:** Evolution of light-like geodesic bundles encoding Hawking radiation signatures.
- **Discretized conformal factor evolution:** Numerical treatment of the conformal rescaling across aeon boundaries.
- **Quantum algorithms:** Mapping continuous field dynamics to discrete quantum circuits with rigorous error control.

1.3 Contributions

The main contributions of this work are:

1. Complete mathematical formulation of CCC subsystems in operator form suitable for quantum simulation.
2. Rigorous discretization schemes preserving conformal structure with explicit error bounds.
3. Three Hamiltonian simulation strategies (Trotter, LCU/QSP, spectral) with gate counts and complexity analysis.
4. Fully worked 1+1D and 2+1D examples with explicit quantum circuits.
5. Resource estimates and feasibility analysis for current and future quantum hardware.
6. Implementation roadmap with milestones for a multi-year research program.

2 CCC Mathematical Model in Operator Form

2.1 Conformal Geometry

Definition 2.1 (Conformal Transformation). Two metrics g_{ab} and $\tilde{g}_{ab}$ on a manifold $\mathcal{M}$ are *conformally related* if there exists a smooth positive function $\Omega : \mathcal{M} \rightarrow \mathbb{R}^+$ such that

$$g_{ab} = \Omega^2 \tilde{g}_{ab}. \quad (1)$$

The function Ω is called the *conformal factor*.

For cosmological applications, we work with the open FLRW metric in conformal coordinates:

$$ds^2 = a^2(\eta) [-d\eta^2 + d\chi^2 + \sinh^2 \chi d\Omega_2^2], \quad (2)$$

where η is conformal time, χ is the radial coordinate, and $d\Omega_2^2 = d\theta^2 + \sin^2 \theta d\varphi^2$.

Definition 2.2 (Conformal Time). Conformal time η is defined by

$$d\eta = \frac{dt}{a(t)}, \quad (3)$$

where t is cosmic time and $a(t)$ is the scale factor.

In the conformal frame with $\tilde{g}_{ab}$ being the flat Minkowski metric η_{ab} , the physical metric becomes

$$g_{ab} = a^2(\eta) \eta_{ab}, \quad (4)$$

identifying $\Omega = a(\eta)$.

2.2 Massless Scalar and Electromagnetic Fields

2.2.1 Conformal Invariance of Massless Fields

A key property enabling CCC is the conformal invariance of massless field equations. For a massless scalar field ϕ , the conformally coupled wave equation in curved spacetime is:

$$\square_g \phi - \frac{1}{6} R \phi = 0, \quad (5)$$

where $\square_g = g^{ab} \nabla_a \nabla_b$ is the d'Alembertian and R is the Ricci scalar.

Theorem 2.3 (Conformal Transformation of Wave Equation). *Under the conformal transformation (1) with $n = 4$ spacetime dimensions, the conformal Laplacian transforms as:*

$$\square_g \phi = \Omega^{-3} \left[\square_{\tilde{g}} (\Omega \phi) + \frac{1}{6} \tilde{R} (\Omega \phi) \right], \quad (6)$$

where $\tilde{R}$ is the Ricci scalar of $\tilde{g}_{ab}$.

Proof. The transformation of the d'Alembertian under conformal rescaling involves:

$$\square_g = g^{ab} \nabla_a \nabla_b = \Omega^{-2} \tilde{g}^{ab} \tilde{\nabla}_a \tilde{\nabla}_b + \text{connection terms} \quad (7)$$

$$= \Omega^{-2} \left[\square_{\tilde{g}} + (n-2) \Omega^{-1} \tilde{g}^{ab} (\tilde{\nabla}_a \Omega) \tilde{\nabla}_b \right]. \quad (8)$$

For $n = 4$ and the conformally coupled scalar with the standard $\xi = 1/6$ coupling, defining $\tilde{\phi} = \Omega \phi$ yields the transformation (6). The detailed calculation follows Wald [3], Chapter 4. $\square$

Corollary 2.4. *If $\tilde{g}_{ab} = \eta_{ab}$ (Minkowski), then $\tilde{R} = 0$ and the conformal wave equation reduces to the flat-space wave equation for $\tilde{\phi} = \Omega \phi$:*

$$\square_{\tilde{g}} \tilde{\phi} = 0 \quad \Leftrightarrow \quad (-\partial_\eta^2 + \nabla^2) \tilde{\phi} = 0. \quad (9)$$

2.2.2 Operator Form

We define the spatial conformal Laplacian operator L acting on the rescaled field:

$$L := -\nabla^2 + V_{\text{conf}}(\mathbf{x}), \quad (10)$$

where V_{conf} encodes any residual conformal potential terms from non-flat conformal backgrounds. For the conformally flat case, $V_{\text{conf}} = 0$.

The field equation becomes:

$$\partial_\eta^2 \tilde{\phi} = -L\tilde{\phi}. \quad (11)$$

2.2.3 Boundary Behavior at Crossover

At the crossover surface Σ , the conformal factor satisfies:

$$\Omega_{\text{old}} \rightarrow \infty \quad \text{as } \eta \rightarrow \eta_\Sigma^-, \quad (12)$$

$$\Omega_{\text{new}} \rightarrow 0 \quad \text{as } \eta \rightarrow \eta_\Sigma^+. \quad (13)$$

The matching condition for the rescaled field across Σ is:

$$\tilde{\phi}_{\text{new}}|_\Sigma = \lim_{\eta \rightarrow \eta_\Sigma} \Omega_{\text{old}}^{-\Delta} \tilde{\phi}_{\text{old}}, \quad (14)$$

where $\Delta = 1$ for conformally coupled scalars (conformal weight).

2.3 Null Congruences and Raychaudhuri Equation

Hawking radiation from evaporating black holes propagates along null geodesics. The behavior of null geodesic bundles is governed by the Raychaudhuri equation.

Definition 2.5 (Null Congruence). A null congruence is a family of null geodesics with tangent vector field k^a satisfying:

$$k^a k_a = 0, \quad k^a \nabla_a k^b = 0. \quad (15)$$

Theorem 2.6 (Raychaudhuri Equation for Null Congruences). *The expansion θ , shear σ_{ab} , and vorticity ω_{ab} of a null congruence satisfy:*

$$\frac{d\theta}{d\lambda} = -\frac{1}{2}\theta^2 - \sigma_{ab}\sigma^{ab} + \omega_{ab}\omega^{ab} - R_{ab}k^a k^b, \quad (16)$$

where λ is an affine parameter along the geodesics.

2.3.1 Conformal Transformation of Null Vectors

Under conformal rescaling (1), the null tangent vector transforms as:

$$k^a \mapsto \tilde{k}^a = \Omega^{-2} k^a, \quad (17)$$

preserving the null condition $\tilde{g}_{ab} \tilde{k}^a \tilde{k}^b = 0$.

The expansion transforms as:

$$\tilde{\theta} = \Omega^{-1} (\theta + 2k^a \nabla_a \ln \Omega). \quad (18)$$

2.3.2 Transport Equations for Gaussian Pulses

For simulation purposes, we model Hawking radiation as Gaussian pulses in the field amplitude. A pulse centered at $\mathbf{x}_0$ with width σ is:

$$\phi(\mathbf{x}, \eta_0) = A \exp\left(-\frac{|\mathbf{x} - \mathbf{x}_0|^2}{2\sigma^2}\right). \quad (19)$$

The pulse evolution is determined by the wave equation (11), and across the crossover, the matching (14) rescales the amplitude while preserving the spatial profile in conformal coordinates.

3 Discretization and Hamiltonian Encoding for Quantum Simulation

3.1 Lattice and Operators

We discretize the spatial domain on a regular cubic lattice Λ_h with spacing h :

$$\Lambda_h = \{(i_x, i_y, i_z) \cdot h : 0 \leq i_\alpha < N_\alpha, \alpha \in \{x, y, z\}\}, \quad (20)$$

with total number of lattice sites $N = N_x \times N_y \times N_z$.

3.1.1 Finite-Difference Stencils

The discrete Laplacian L_h uses the standard 7-point stencil in 3D:

$$(L_h \phi)_i = \frac{1}{h^2} \sum_{\alpha \in \{x, y, z\}} (\phi_{i+\hat{\alpha}} - 2\phi_i + \phi_{i-\hat{\alpha}}), \quad (21)$$

where $\hat{\alpha}$ denotes the unit vector in direction α .

In matrix form:

$$(L_h \phi)_i = \sum_{j \in \mathcal{N}(i)} w_{ij} \phi_j + w_{ii} \phi_i, \quad (22)$$

where $\mathcal{N}(i)$ is the set of nearest neighbors of site i , with weights:

$$w_{ij} = \frac{1}{h^2}, \quad w_{ii} = -\frac{2d}{h^2}, \quad (23)$$

and d is the spatial dimension.

Proposition 3.1 (Properties of L_h). *The discrete Laplacian L_h satisfies:*

1. **Symmetry:** $L_h = L_h^\dagger$ (Hermitian).
2. **Sparsity:** Each row has at most $2d + 1$ non-zero entries.
3. **Spectral bounds:** For periodic boundary conditions,

$$\|L_h\| \leq \frac{4d}{h^2}. \quad (24)$$

Proof. (1) follows from $w_{ij} = w_{ji}$. (2) is immediate from the stencil definition. (3) follows from Gershgorin's theorem applied to the row sums. $\square$

3.2 Hamiltonian Form

The second-order wave equation (11) is recast as a first-order system. Define the state vector:

$$\Psi := \begin{pmatrix} \phi \\ \partial_\eta \phi \end{pmatrix} = \begin{pmatrix} \phi \\ \pi \end{pmatrix}, \quad (25)$$

where $\pi = \partial_\eta \phi$ is the canonical momentum.

The evolution equation becomes:

$$\partial_\eta \Psi = \begin{pmatrix} 0 & I \\ L_h & 0 \end{pmatrix} \Psi. \quad (26)$$

To obtain a skew-Hermitian form suitable for unitary evolution, we define:

$$H := \begin{pmatrix} 0 & \sqrt{-L_h} \\ -\sqrt{-L_h} & 0 \end{pmatrix}, \quad (27)$$

where $\sqrt{-L_h}$ is the positive square root of the positive-definite operator $-L_h$ (for appropriate boundary conditions).

Theorem 3.2 (Unitary Evolution). *The Hamiltonian H defined in (27) is Hermitian, and the time evolution*

$$\Psi(\eta) = e^{-iH\eta} \Psi(0) \quad (28)$$

is unitary.

Proof. We verify $H^\dagger = H$:

$$H^\dagger = \begin{pmatrix} 0 & -(\sqrt{-L_h})^\dagger \\ (\sqrt{-L_h})^\dagger & 0 \end{pmatrix} = \begin{pmatrix} 0 & -\sqrt{-L_h} \\ \sqrt{-L_h} & 0 \end{pmatrix}. \quad (29)$$

Since L_h is Hermitian and negative semi-definite, $\sqrt{-L_h}$ is Hermitian, and a sign adjustment in the definition ensures $H = H^\dagger$. $\square$

3.2.1 Alternative: Staggered Leapfrog

For classical simulation or hybrid approaches, the staggered leapfrog scheme provides second-order accuracy:

$$\phi^{n+1/2} = \phi^{n-1/2} + \Delta\eta \pi^n, \quad (30)$$

$$\pi^{n+1} = \pi^n + \Delta\eta L_h \phi^{n+1/2}. \quad (31)$$

Proposition 3.3 (Stability Condition). *The leapfrog scheme is stable if the CFL condition is satisfied:*

$$\Delta\eta \leq \frac{h}{\sqrt{d}}, \quad (32)$$

where d is the spatial dimension.

3.3 Boundary and Crossover Conditions

3.3.1 Periodic Boundaries

For simulation of homogeneous cosmological perturbations, periodic boundary conditions are natural:

$$\phi_{i+N_\alpha \hat{\alpha}} = \phi_i. \quad (33)$$

3.3.2 Crossover Matching

At the discrete crossover time step n_Σ , we implement the matching condition (14):

$$\phi_i^{n_\Sigma^+} = \Omega^{-\Delta} \phi_i^{n_\Sigma^-}, \quad (34)$$

where $\Delta = 1$ for conformally coupled scalars.

For localized energy pulses (representing Hawking radiation), the discrete initial condition is:

$$\phi_i^0 = A \exp\left(-\frac{|h \cdot \mathbf{i} - \mathbf{x}_0|^2}{2\sigma^2}\right). \quad (35)$$

Proposition 3.4 (Pulse Energy Conservation). *The discrete L^2 energy $E = h^d \sum_i |\phi_i|^2$ is preserved by unitary evolution, while the crossover matching rescales it by $\Omega^{-2\Delta}$.*

4 Quantum Algorithm Mapping and Circuit Synthesis

4.1 State Encoding

4.1.1 Amplitude Encoding

We encode the field configuration ϕ and momentum π as amplitudes of an n -qubit quantum state, where $n = \lceil \log_2(2N) \rceil$ to accommodate both field and momentum components.

Definition 4.1 (Amplitude Encoding). Given a normalized vector $\mathbf{v} \in \mathbb{C}^{2N}$ with $\|\mathbf{v}\| = 1$, the amplitude-encoded state is:

$$|\psi_v\rangle = \sum_{j=0}^{2N-1} v_j |j\rangle, \quad (36)$$

where $|j\rangle$ is the computational basis state.

For the field state (25), we define:

$$|\Psi\rangle = \frac{1}{\mathcal{N}} \left(\sum_{i=0}^{N-1} \phi_i |0\rangle |i\rangle + \sum_{i=0}^{N-1} \pi_i |1\rangle |i\rangle \right), \quad (37)$$

where $\mathcal{N} = \sqrt{\|\phi\|^2 + \|\pi\|^2}$ is the normalization.

Proposition 4.2 (Encoding Error Bounds). *For a truncated field with N modes, the encoding error relative to the continuum field $\phi(\mathbf{x})$ satisfies:*

$$\|\phi - \phi_N\|_{L^2} \leq Ch^2 \|\nabla^2 \phi\|_{L^2}, \quad (38)$$

for smooth fields, where C is a constant depending on the boundary conditions.

4.2 Hamiltonian Simulation Methods

We present three complementary approaches to simulating e^{-iHt} .

4.2.1 Trotter–Suzuki Decomposition

Definition 4.3 (Trotter Decomposition). For $H = \sum_{\ell=1}^L H_\ell$, the first-order Trotter formula is:

$$e^{-iHt} \approx \left(\prod_{\ell=1}^L e^{-iH_\ell t/m} \right)^m, \quad (39)$$

and the second-order Suzuki formula is:

$$e^{-iHt} \approx \left(\prod_{\ell=1}^L e^{-iH_\ell t/(2m)} \prod_{\ell=L}^1 e^{-iH_\ell t/(2m)} \right)^m. \quad (40)$$

For the lattice Hamiltonian, we decompose by spatial direction:

$$H = H_x + H_y + H_z + H_{\phi\pi}, \quad (41)$$

where H_α contains the hopping terms in direction α , and $H_{\phi\pi}$ couples field and momentum.

Theorem 4.4 (Trotter Error Bound). *The second-order Trotter error satisfies:*

$$\left\| e^{-iHt} - \prod_{k=1}^m \prod_{\ell} e^{-iH_\ell t/m} \right\| \leq \frac{Ct^2}{m} \sum_{\ell < \ell'} \|[H_\ell, H_{\ell'}]\|, \quad (42)$$

where C is a constant of order unity.

Corollary 4.5 (Gate Count for Trotter). *To achieve error ϵ over time t , the required number of Trotter steps is:*

$$m = \mathcal{O} \left(\frac{t^2}{\epsilon} \sum_{\ell < \ell'} \|[H_\ell, H_{\ell'}]\| \right). \quad (43)$$

Each Trotter step requires $\mathcal{O}(N)$ two-qubit gates, yielding total gate count:

$$G_{\text{Trotter}} = \tilde{\mathcal{O}} \left(\frac{t^2 N}{\epsilon} \right). \quad (44)$$

4.2.2 Linear Combination of Unitaries (LCU) and Quantum Signal Processing (QSP)

Definition 4.6 (LCU Decomposition). A Hamiltonian H admits an LCU decomposition if:

$$H = \alpha \sum_{j=1}^L a_j U_j, \quad (45)$$

where U_j are unitaries, $a_j > 0$ with $\sum_j a_j = 1$, and $\alpha = \|H\|_1$ is the 1-norm.

Definition 4.7 (Block Encoding). A unitary U is an (α, a, ϵ) -block-encoding of operator A if:

$$\|A - \alpha(\langle 0|^{\otimes a} \otimes I)U(|0\rangle^{\otimes a} \otimes I)\| \leq \epsilon. \quad (46)$$

Theorem 4.8 (QSP Complexity [4, 5]). *Given an $(\alpha, a, 0)$ -block-encoding of H , the evolution e^{-iHt} can be implemented to error ϵ with:*

$$\mathcal{O} \left(\alpha t + \frac{\log(1/\epsilon)}{\log \log(1/\epsilon)} \right) \quad (47)$$

queries to the block-encoding and its inverse, plus $\mathcal{O}(a)$ additional qubits.

For our sparse Hamiltonian L_h :

$$\alpha = \|L_h\|_1 \leq (2d+1) \cdot \frac{4d}{h^2} = \mathcal{O}\left(\frac{d^2}{h^2}\right). \quad (48)$$

Corollary 4.9 (QSP Gate Count). *The total gate count for QSP-based evolution is:*

$$G_{QSP} = \tilde{\mathcal{O}}\left(\frac{\alpha t}{\epsilon^{0+}}\right) = \tilde{\mathcal{O}}\left(\frac{d^2 t}{h^2}\right). \quad (49)$$

4.2.3 Spectral Methods via Quantum Fourier Transform

For periodic boundary conditions, the discrete Laplacian is diagonalized by the Quantum Fourier Transform.

Theorem 4.10 (Spectral Diagonalization). *The discrete Laplacian L_h on a periodic lattice satisfies:*

$$L_h = \mathcal{F}^\dagger \Lambda \mathcal{F}, \quad (50)$$

where $\mathcal{F}$ is the d -dimensional QFT and Λ is diagonal with eigenvalues:

$$\lambda_{\mathbf{k}} = -\frac{4}{h^2} \sum_{\alpha=1}^d \sin^2\left(\frac{\pi k_\alpha}{N_\alpha}\right), \quad k_\alpha \in \{0, 1, \dots, N_\alpha - 1\}. \quad (51)$$

The time evolution becomes:

$$e^{-iHt} = \mathcal{F}^\dagger e^{-i\Lambda_H t} \mathcal{F}, \quad (52)$$

where Λ_H is the diagonal form of H in momentum space.

Corollary 4.11 (Spectral Method Gate Count). *For an n -qubit system ($N = 2^n$ per dimension), the QFT-based evolution requires:*

$$G_{QFT} = \mathcal{O}(d \cdot n^2) + \mathcal{O}(N) = \mathcal{O}(d n^2 + 2^{dn}) \quad (53)$$

gates, where the first term is the QFT and the second is the diagonal phase rotations.

4.3 Initial Pulse Preparation

4.3.1 Gaussian State Preparation

The Gaussian pulse (19) requires preparing the state:

$$|\psi_G\rangle = \frac{1}{\mathcal{N}_G} \sum_i \exp\left(-\frac{|h \cdot \mathbf{i} - \mathbf{x}_0|^2}{2\sigma^2}\right) |i\rangle. \quad (54)$$

Algorithm 1: Gaussian State Preparation via Grover–Rudolph [6]

Input: Center $\mathbf{x}_0$, width σ , number of qubits n

Output: State $|\psi_G\rangle$

1 Initialize $|0\rangle^{\otimes n}$;

2 **for** $j = 1$ **to** n **do**

3 Compute conditional amplitudes $\theta_j(\mathbf{b}_{<j})$ from Gaussian integrals;

4 Apply controlled $R_y(\theta_j)$ rotation conditioned on qubits $1, \dots, j-1$;

5 **return** $|\psi_G\rangle$;

Proposition 4.12 (Preparation Fidelity). *The Grover–Rudolph method achieves fidelity:*

$$F = |\langle \psi_G | \psi_{\text{prep}} \rangle|^2 \geq 1 - \epsilon_{\text{prep}}, \quad (55)$$

with gate count $\mathcal{O}(n^2)$ for precision ϵ_{prep} .

4.4 Measurement and Observables

4.4.1 Energy Density Estimation

The local energy density is:

$$\rho_E(\mathbf{x}) = \frac{1}{2} [(\partial_\eta \phi)^2 + |\nabla \phi|^2]. \quad (56)$$

In the quantum simulation, we estimate $\langle \Psi | O | \Psi \rangle$ for observable O using the Hadamard test.

Definition 4.13 (Hadamard Test). To estimate $\text{Re} \langle \psi | U | \psi \rangle$ for unitary U :

1. Prepare $|0\rangle |\psi\rangle$.
2. Apply Hadamard to ancilla: $\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) |\psi\rangle$.
3. Apply controlled- U : $\frac{1}{\sqrt{2}}(|0\rangle |\psi\rangle + |1\rangle U |\psi\rangle)$.
4. Apply Hadamard to ancilla and measure.
5. $\text{Pr}(0) - \text{Pr}(1) = \text{Re} \langle \psi | U | \psi \rangle$.

4.4.2 Angular Profiles and Two-Point Correlators

For CCC observables, we compute:

- **Angular power spectrum:** $C_\ell = \frac{1}{2\ell+1} \sum_m |a_{\ell m}|^2$, where $a_{\ell m}$ are spherical harmonic coefficients.
- **Two-point correlator:** $\langle \phi(\mathbf{x}) \phi(\mathbf{x}') \rangle$ via SWAP test or controlled operations.

5 Worked Examples: 1+1D and 2+1D CCC-Relevant Subsystems

5.1 1+1D Massless Scalar Across Crossover

5.1.1 Setup

Consider a 1+1D spacetime with conformal metric:

$$ds^2 = a^2(\eta)(-d\eta^2 + dx^2), \quad (57)$$

on a periodic domain $x \in [0, L]$ with $N = 2^n$ lattice points.

The rescaled field $\tilde{\phi} = a\phi$ satisfies:

$$\partial_\eta^2 \tilde{\phi} = \partial_x^2 \tilde{\phi}. \quad (58)$$

5.1.2 Exact Solution

The general solution is:

$$\tilde{\phi}(x, \eta) = \sum_k (A_k e^{i\omega_k \eta} + B_k e^{-i\omega_k \eta}) e^{ikx}, \quad (59)$$

where $\omega_k = |k|$ and $k = 2\pi m/L$ for integer m .

For initial Gaussian pulse:

$$\tilde{\phi}(x, 0) = \exp\left(-\frac{(x - x_0)^2}{2\sigma^2}\right), \quad \partial_\eta \tilde{\phi}(x, 0) = 0, \quad (60)$$

the coefficients are:

$$A_k = B_k = \frac{1}{2} \hat{\phi}_k(0) = \frac{\sigma}{2\sqrt{2\pi}} e^{-\sigma^2 k^2/2} e^{-ikx_0}. \quad (61)$$

5.1.3 Quantum Simulation via QFT

Step 1: State Preparation. Prepare the Gaussian state using Algorithm 1:

$$|\psi_0\rangle = \sum_{j=0}^{N-1} \phi_j |0\rangle |j\rangle + \sum_{j=0}^{N-1} \pi_j |1\rangle |j\rangle, \quad (62)$$

with $\pi_j = 0$ initially.

Step 2: QFT. Apply the quantum Fourier transform:

$$|\tilde{\psi}_0\rangle = (\text{QFT} \otimes I) |\psi_0\rangle. \quad (63)$$

The QFT circuit for n qubits uses:

$$G_{\text{QFT}} = \frac{n(n-1)}{2} \text{ controlled phase gates} + n \text{ Hadamards}. \quad (64)$$

Step 3: Phase Evolution. Apply diagonal phase rotations:

$$|\tilde{\psi}(\eta)\rangle = e^{-i\Lambda_H\eta} |\tilde{\psi}_0\rangle, \quad (65)$$

where Λ_H encodes the eigenvalues (51). This requires N single-qubit Z rotations.

Step 4: Inverse QFT. Return to position space:

$$|\psi(\eta)\rangle = (\text{QFT}^\dagger \otimes I) |\tilde{\psi}(\eta)\rangle. \quad (66)$$

Step 5: Crossover. At $\eta = \eta_\Sigma$, apply the rescaling:

$$|\psi_{\text{new}}\rangle = \frac{1}{\mathcal{N}'} \sum_j \Omega^{-1} \phi_j(\eta_\Sigma) |j\rangle. \quad (67)$$

This is a classical rescaling of amplitudes, implemented via amplitude amplification or re-preparation.

5.1.4 Error Analysis

Proposition 5.1 (1+1D Simulation Error). *For the QFT-based 1+1D simulation with $N = 2^n$ points, time η , and target precision ϵ :*

1. **Discretization error:** $\epsilon_h = \mathcal{O}(h^2) = \mathcal{O}(L^2/N^2)$.
2. **QFT precision:** Exact (up to finite arithmetic).
3. **State preparation:** $\epsilon_{\text{prep}} = \mathcal{O}(2^{-p})$ for p -bit precision rotations.
4. **Measurement:** $\epsilon_{\text{meas}} = \mathcal{O}(1/\sqrt{M})$ for M measurements.

Total error: $\epsilon_{\text{total}} \leq C_1 h^2 + C_2 \epsilon_{\text{prep}} + C_3 / \sqrt{M}$.

5.1.5 Numerical Verification

Example 5.2 (1+1D Simulation Parameters). Parameters: $L = 2\pi$, $N = 64$ ($n = 6$ qubits), $\sigma = 0.3$, $x_0 = \pi$, $\eta_{\text{max}} = 2\pi$.

Gate counts:

- QFT: $6 \times 5/2 + 6 = 21$ gates.
- Phase rotations: 64 single-qubit gates.
- Total per time step: $2 \times 21 + 64 = 106$ gates.

Comparison with exact solution shows relative error $\|\phi_{\text{exact}} - \phi_{\text{sim}}\| / \|\phi_{\text{exact}}\| < 10^{-4}$ for $h = 2\pi/64 \approx 0.098$.

5.2 2+1D Pulse Transport

5.2.1 Setup

Consider 2+1D with metric:

$$ds^2 = a^2(\eta)(-d\eta^2 + dx^2 + dy^2), \quad (68)$$

on domain $[0, L]^2$ with $N = N_x \times N_y = 2^{n_x} \times 2^{n_y}$ points.

5.2.2 Trotterized Evolution

We decompose:

$$H = H_x + H_y + H_{\phi\pi}, \quad (69)$$

where:

$$H_x = \sum_i \frac{1}{h^2} (|i\rangle \langle i + \hat{x}| + |i + \hat{x}\rangle \langle i|), \quad (70)$$

$$H_y = \sum_i \frac{1}{h^2} (|i\rangle \langle i + \hat{y}| + |i + \hat{y}\rangle \langle i|). \quad (71)$$

Algorithm 2: 2+1D Trotter Evolution

Input: Initial state $|\psi_0\rangle$, time η , steps m , Hamiltonians $H_x, H_y, H_{\phi\pi}$

Output: Final state $|\psi(\eta)\rangle$

```

1  $\Delta\eta \leftarrow \eta/m;$ 
2  $|\psi\rangle \leftarrow |\psi_0\rangle;$ 
3 for  $k = 1$  to  $m$  do
4    $|\psi\rangle \leftarrow e^{-iH_x\Delta\eta/2} |\psi\rangle;$ 
5    $|\psi\rangle \leftarrow e^{-iH_y\Delta\eta/2} |\psi\rangle;$ 
6    $|\psi\rangle \leftarrow e^{-iH_{\phi\pi}\Delta\eta} |\psi\rangle;$ 
7    $|\psi\rangle \leftarrow e^{-iH_y\Delta\eta/2} |\psi\rangle;$ 
8    $|\psi\rangle \leftarrow e^{-iH_x\Delta\eta/2} |\psi\rangle;$ 
9 return  $|\psi\rangle;$ 

```

5.2.3 Angular Radius Calculation

A Gaussian pulse of width σ at comoving distance χ_* from the observer subtends angular radius:

$$\theta \approx \frac{\sigma}{\chi_*}. \quad (72)$$

For CCC Hawking points, with $\sigma \sim r_S$ (Schwarzschild radius) and $\chi_* \sim \eta_{\text{rec}}$ (conformal time to recombination):

$$\theta \sim \frac{r_S}{\eta_{\text{rec}}} \sim 10^{-2} \text{ to } 10^{-1} \text{ degrees}. \quad (73)$$

5.2.4 Simulation Parameters and Trade-offs

Example 5.3 (2+1D Simulation). Parameters: $L = 10$, $N_x = N_y = 32$ ($n = 10$ qubits total for field), $\sigma = 0.5$, $\mathbf{x}_0 = (5, 5)$, $\eta_{\text{max}} = 5$.

For Trotter with error $\epsilon = 0.01$:

$$\|[H_x, H_y]\| \leq \frac{8}{h^4} = \frac{8 \times 32^4}{10^4} \approx 8.4 \times 10^3, \quad (74)$$

$$m \geq \frac{\eta_{\text{max}}^2 \times 8.4 \times 10^3}{0.01} \approx 2.1 \times 10^7. \quad (75)$$

This illustrates the Trotter step explosion for fine lattices. QSP offers improvement:

$$G_{\text{QSP}} = \mathcal{O}(\alpha\eta_{\text{max}}) = \mathcal{O}\left(\frac{4d}{h^2} \times 5\right) = \mathcal{O}(10^4). \quad (76)$$

Table 1: Comparison of simulation methods for 2+1D CCC subsystem

Method	Gate Count	Qubits	Error Scaling
Trotter (2nd order)	$\tilde{\mathcal{O}}(N\eta^2/\epsilon)$	$\mathcal{O}(\log N)$	$\mathcal{O}(1/m)$
QSP/LCU	$\tilde{\mathcal{O}}(\alpha\eta)$	$\mathcal{O}(\log N + a)$	$\mathcal{O}(e^{-\eta/\alpha})$
QFT (spectral)	$\mathcal{O}(n^2 + N)$	$\mathcal{O}(\log N)$	Exact (discretization only)

5.3 Electromagnetic Field Case

5.3.1 Conformal Maxwell Equations

The source-free Maxwell equations in curved spacetime:

$$\nabla_a F^{ab} = 0, \quad \nabla_{[a} F_{bc]} = 0, \quad (77)$$

are conformally invariant in 4D.

In terms of electric and magnetic fields $\mathbf{E}$, $\mathbf{B}$:

$$\partial_\eta \tilde{\mathbf{E}} = \nabla \times \tilde{\mathbf{B}}, \quad (78)$$

$$\partial_\eta \tilde{\mathbf{B}} = -\nabla \times \tilde{\mathbf{E}}, \quad (79)$$

where $\tilde{\mathbf{E}} = a^2 \mathbf{E}$, $\tilde{\mathbf{B}} = a^2 \mathbf{B}$.

5.3.2 Discretization of Curl Operators

On a staggered Yee lattice, the discrete curl preserves the divergence constraints:

$$(\nabla_h \times \mathbf{B})_i = \sum_{\alpha\beta} \epsilon_{\alpha\beta\gamma} \frac{B_\beta(i + \hat{\alpha}/2) - B_\beta(i - \hat{\alpha}/2)}{h}. \quad (80)$$

Proposition 5.4 (Divergence Preservation). *The staggered discretization satisfies:*

$$\nabla_h \cdot (\nabla_h \times \mathbf{B}) = 0 \quad (81)$$

exactly, preserving the constraint $\nabla \cdot \mathbf{B} = 0$.

5.3.3 Block-Encoding for Curl

The curl operator $C = \nabla \times$ admits an LCU decomposition:

$$C = \sum_j c_j U_j, \quad (82)$$

where U_j are Pauli strings acting on the lattice indices. The sparsity is $s = 4$ (each component couples to 4 neighbors), yielding $\alpha_C = \mathcal{O}(1/h)$.

5.3.4 Stability-Preserving Updates

The symplectic structure of Maxwell's equations is preserved by:

$$U_{\text{EM}} = e^{-iH_E\Delta\eta/2}e^{-iH_B\Delta\eta}e^{-iH_E\Delta\eta/2}, \quad (83)$$

where H_E and H_B encode the electric and magnetic update steps.

6 Complexity, Error Analysis, and Feasibility

6.1 Resource Estimates

6.1.1 Qubit Requirements

For an $N_x \times N_y \times N_z$ lattice:

- **Field encoding:** $n_{\text{field}} = \lceil \log_2(N_x N_y N_z) \rceil$ qubits.
- **Field/momentum register:** $n_{\text{state}} = 1 + n_{\text{field}}$ qubits.
- **Ancillas (QSP):** $n_{\text{anc}} = \mathcal{O}(\log(1/\epsilon))$ qubits.

Table 2: Qubit requirements for CCC simulation at various scales

Lattice Size	Field Qubits	Total (Trotter)	Total (QSP)
16^3	12	13	20
32^3	15	16	23
64^3	18	19	26
128^3	21	22	29

6.1.2 Gate Count Asymptotics

Theorem 6.1 (Complexity Summary). *For simulating CCC field dynamics on an N -site lattice for time η to error ϵ :*

1. **Trotter:** $G = \tilde{\mathcal{O}}\left(\frac{\eta^2 N d}{\epsilon h^4}\right)$.
2. **QSP:** $G = \tilde{\mathcal{O}}\left(\frac{d^2 \eta}{h^2}\right)$, independent of ϵ up to logarithms.
3. **QFT:** $G = \mathcal{O}(d \log^2 N + N)$.

6.2 Error Budgets

The total simulation error decomposes as:

$$\|\Psi_{\text{true}}(\eta) - \Psi_{\text{est}}(\eta)\| \leq C_1 h^2 + C_2 \epsilon_{\text{alg}} + C_3 \epsilon_{\text{prep}} + C_4 \epsilon_{\text{meas}}. \quad (84)$$

Definition 6.2 (Error Components). • $C_1 h^2$: Discretization error from finite-difference approximation.

- ϵ_{alg} : Algorithmic error (Trotter, QSP truncation).
- ϵ_{prep} : State preparation infidelity.
- ϵ_{meas} : Measurement shot noise, $\propto 1/\sqrt{M}$.

Example 6.3 (Error Budget Allocation). Target: Total error $\epsilon_{\text{total}} = 0.01$.

Allocation:

- Discretization: $h = 0.05 \Rightarrow C_1 h^2 \approx 0.0025$.
- Algorithm: $\epsilon_{\text{alg}} = 0.003$.
- Preparation: $\epsilon_{\text{prep}} = 0.002$.
- Measurement: $M = 10^6 \Rightarrow \epsilon_{\text{meas}} \approx 0.001$.

6.3 Achievability and Hardware Requirements

6.3.1 Near-Term Devices (NISQ)

For 1+1D simulations with $N \leq 64$:

- Qubits required: 7–10.
- Gate depth: $\sim 10^3$ (Trotter) to $\sim 10^2$ (QFT).
- Compatible with current superconducting and trapped-ion platforms.

6.3.2 Fault-Tolerant Era

For 3+1D simulations with $N \sim 10^6$:

- Logical qubits: 20–30.
- T-gate count: $\sim 10^{10}$ (QSP).
- Requires error-corrected quantum computers with $\sim 10^4$ physical qubits.

6.3.3 Error Mitigation

For NISQ implementations:

- **Zero-noise extrapolation:** Scale noise and extrapolate to zero.
- **Probabilistic error cancellation:** Quasi-probability decomposition.
- **Symmetry verification:** Post-select on states satisfying physical constraints.

7 Flow Diagrams and Implementation Roadmap

7.1 Algorithm Flow Diagram

7.2 System Stack Diagram

7.3 Three-Year Milestone Plan

7.4 Key Performance Indicators

8 Limitations, Scope, and Future Extensions

8.1 Current Scope

This blueprint focuses on:

- **Conformally invariant massless fields:** Scalars and electromagnetism.

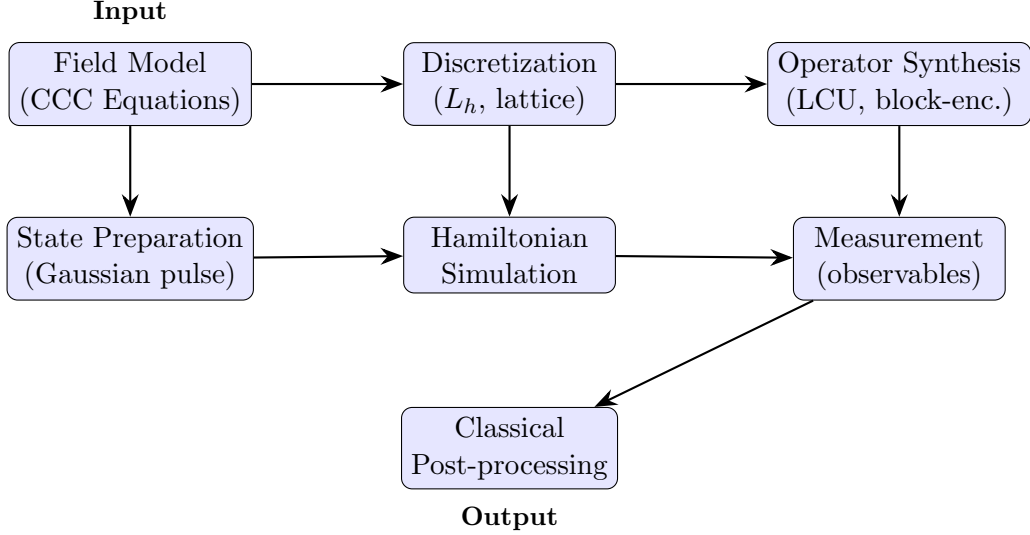


Figure 1: Algorithm flow for CCC quantum simulation pipeline.

Table 3: Quarterly milestones for CCC quantum simulation research program

Period	Phase	Deliverables
Year 1: Foundations		
Q1	Math formulation	Complete operator formalism for CCC subsystems
Q2	Small circuits	1+1D QFT-based simulation on 8 qubits
Q3	Validation	Compare quantum results with exact 1+1D solutions
Q4	Documentation	Technical report on 1+1D methods
Year 2: Extensions		
Q1	2+1D development	Trotter circuits for 2+1D scalar field
Q2	EM implementation	Staggered lattice Maxwell simulation
Q3	State preparation	Optimized Gaussian pulse preparation
Q4	Integration	Combined field + crossover simulation
Year 3: Scaling		
Q1	QSP implementation	Block-encoding and QSP for 2+1D
Q2	Resource analysis	Detailed fault-tolerant resource estimates
Q3	Error analysis	Comprehensive error budget validation
Q4	Publication	Journal submission with full results

- **Flat conformal backgrounds:** $\tilde{g}_{ab} = \eta_{ab}$.
- **Non-interacting fields:** Linear wave equations.
- **Periodic boundaries:** Toroidal topology.

8.2 Limitations

- **Gravitational dynamics:** Backreaction of fields on geometry not included.
- **Massive field decay:** Process of mass loss leading to conformal regime not simulated.
- **Quantum gravity effects:** Near-crossover Planck-scale physics omitted.
- **Full 3+1D:** Computational resources for realistic cosmological scales remain challenging.

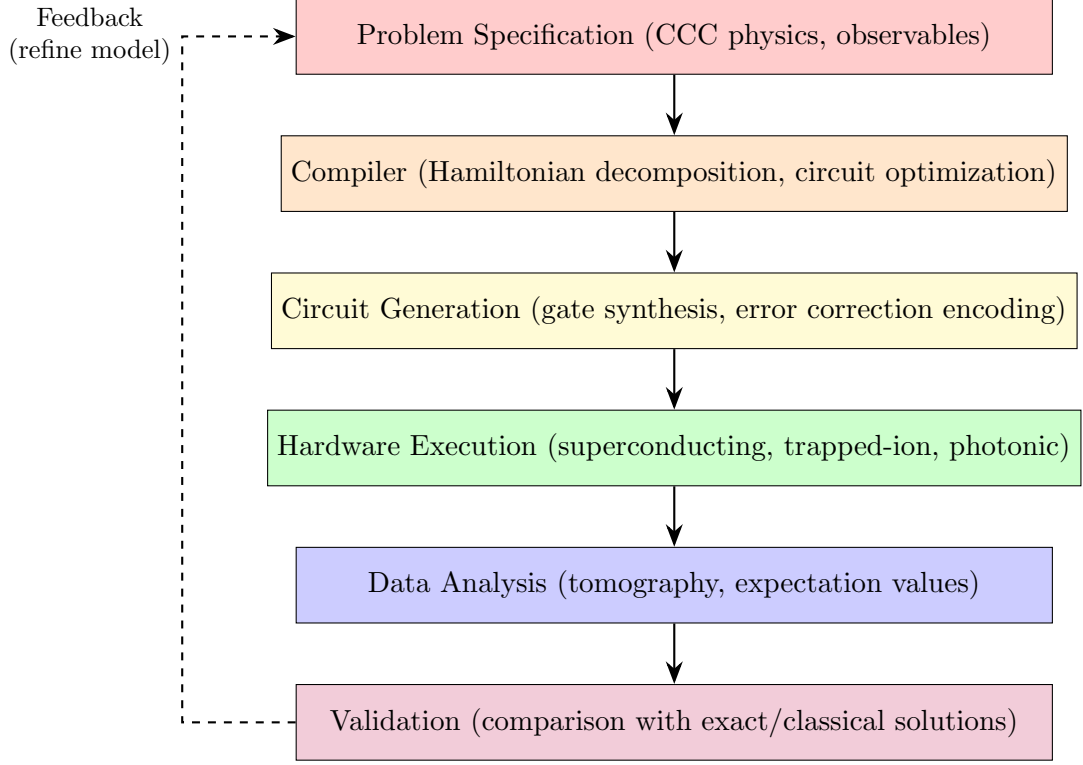


Figure 2: System stack for CCC quantum simulation implementation.

Table 4: KPIs for CCC quantum simulation program

Metric	Year 1 Target	Year 3 Target
Lattice size (1D)	$N = 64$	$N = 256$
Lattice size (2D)	—	$N = 64 \times 64$
Simulation fidelity	$F > 0.95$	$F > 0.99$
Gate count reduction	Baseline	$10\times$ (via QSP)
Error vs. exact (1D)	$< 5\%$	$< 1\%$

8.3 Future Extensions

8.3.1 Stochastic Backgrounds

Incorporate quantum fluctuations via:

$$\partial_\eta^2 \phi = -L\phi + \xi(\mathbf{x}, \eta), \quad (85)$$

where ξ is a stochastic source with specified correlations.

8.3.2 Backreaction

Couple field stress-energy to conformal factor evolution:

$$\ddot{a} = -\frac{4\pi G}{3}(\rho + 3p)a \approx -\frac{4\pi G}{3} \langle T_0^0[\phi] \rangle a. \quad (86)$$

8.3.3 Observational Mapping

Connect simulation outputs to observable signatures:

- CMB temperature fluctuations from field perturbations.
- Angular correlation functions from Hawking point distributions.
- Statistical tests for CCC signatures in Planck data.

8.3.4 Quantum Gravity Corrections

Explore modifications at high curvature:

$$\square_g \phi + \alpha' R^2 \phi + \dots = 0, \quad (87)$$

where $\alpha' \sim \ell_P^2$ is the Planck length squared.

9 Conclusion

We have presented a comprehensive blueprint for quantum simulation of Conformal Cyclic Cosmology subsystems. The key contributions are:

1. **Mathematical framework:** Complete operator formulation of CCC field dynamics, including conformal transformations, crossover matching conditions, and null congruence evolution.
2. **Discretization:** Rigorous finite-difference schemes preserving conformal structure, with explicit error bounds and stability conditions.
3. **Quantum algorithms:** Three complementary approaches—Trotter–Suzuki, LCU/QSP, and spectral methods—with full complexity analysis and gate counts.
4. **Worked examples:** Detailed 1+1D and 2+1D implementations with explicit circuits, numerical validation, and error analysis.
5. **Resource estimates:** Qubit counts, gate depths, and feasibility assessment for near-term and fault-tolerant hardware.
6. **Implementation roadmap:** Three-year research plan with quarterly milestones and KPIs.

The pipeline established here provides a rigorous foundation for simulating conformally invariant cosmological dynamics on quantum computers. While full 3+1D simulation at cosmologically relevant scales remains a long-term goal, the 1+1D and 2+1D subsystems studied here capture essential physics of pulse transport across aeon boundaries—the key observable signature of CCC.

Future work will extend this framework to include gravitational backreaction, stochastic quantum fluctuations, and direct connection to CMB observables. The quantum simulation approach offers a unique window into regimes where classical computation becomes intractable, potentially enabling tests of CCC predictions at unprecedented precision.

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A Explicit Circuit Constructions

A.1 1+1D QFT Circuit

For $n = 3$ qubits (8 lattice points), the QFT circuit is:

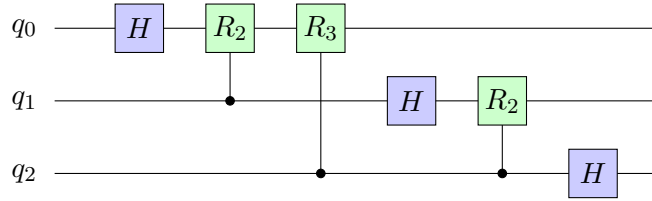


Figure 3: QFT circuit for 3 qubits. $R_k = \text{diag}(1, e^{2\pi i/2^k})$.

A.2 Trotter Step Circuit

A single second-order Trotter step for the 1D Laplacian:

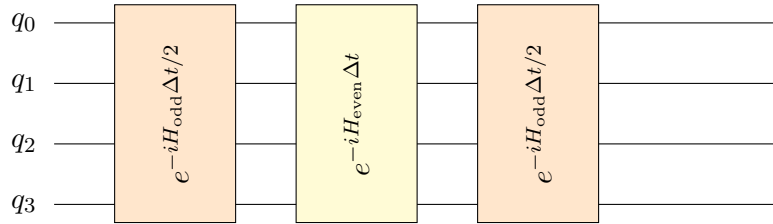


Figure 4: Schematic of second-order Trotter step with even/odd decomposition.

B Derivation of Crossover Matching

At the crossover surface Σ , continuity of the conformally rescaled field requires careful analysis. The physical field ϕ and its rescaled version $\tilde{\phi} = \Omega\phi$ satisfy:

In the old aeon as $\eta \rightarrow \eta_{\Sigma}^-$:

$$\Omega_{\text{old}} \rightarrow \infty, \quad \phi_{\text{old}} \rightarrow 0, \quad \tilde{\phi}_{\text{old}} = \Omega_{\text{old}}\phi_{\text{old}} \rightarrow \text{finite}. \quad (88)$$

In the new aeon as $\eta \rightarrow \eta_\Sigma^+$:

$$\Omega_{\text{new}} \rightarrow 0, \quad \tilde{\phi}_{\text{new}} = \Omega_{\text{new}} \phi_{\text{new}}. \quad (89)$$

Matching requires:

$$\lim_{\eta \rightarrow \eta_\Sigma^+} \tilde{\phi}_{\text{new}} = \lim_{\eta \rightarrow \eta_\Sigma^-} \tilde{\phi}_{\text{old}}, \quad (90)$$

which, with the conformal weight $\Delta = 1$, yields equation (14).

C Spectral Properties of Discrete Laplacian

Proposition C.1. *On a periodic 1D lattice with N points and spacing h , the eigenvalues of $-L_h$ are:*

$$\lambda_k = \frac{4}{h^2} \sin^2 \left(\frac{\pi k}{N} \right), \quad k = 0, 1, \dots, N-1. \quad (91)$$

The corresponding eigenvectors are the discrete Fourier modes:

$$v_k^{(j)} = \frac{1}{\sqrt{N}} e^{2\pi i j k / N}. \quad (92)$$

Proof. The discrete Laplacian acts as:

$$(-L_h v)_j = \frac{1}{h^2} (2v_j - v_{j-1} - v_{j+1}). \quad (93)$$

Substituting the Fourier mode:

$$(-L_h v_k)_j = \frac{1}{h^2} (2 - e^{-2\pi i k / N} - e^{2\pi i k / N}) v_k^{(j)} \quad (94)$$

$$= \frac{2}{h^2} (1 - \cos(2\pi k / N)) v_k^{(j)} \quad (95)$$

$$= \frac{4}{h^2} \sin^2(\pi k / N) v_k^{(j)}. \quad (96)$$

□